



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

TECHNICAL PRESENTATION

CSA1402-COMPILER DESIGN

TOPIC: "PHASES OF A COMPILER"

Guided By
Dr. N.DEEPA

Presented By
Hari vishnu N – 192524319

INTRODUCTION TO COMPILER

❖ **What is a Compiler?**

A compiler is a software program that translates a program written in a high-level language (such as C or Java) into machine language that a computer can execute.

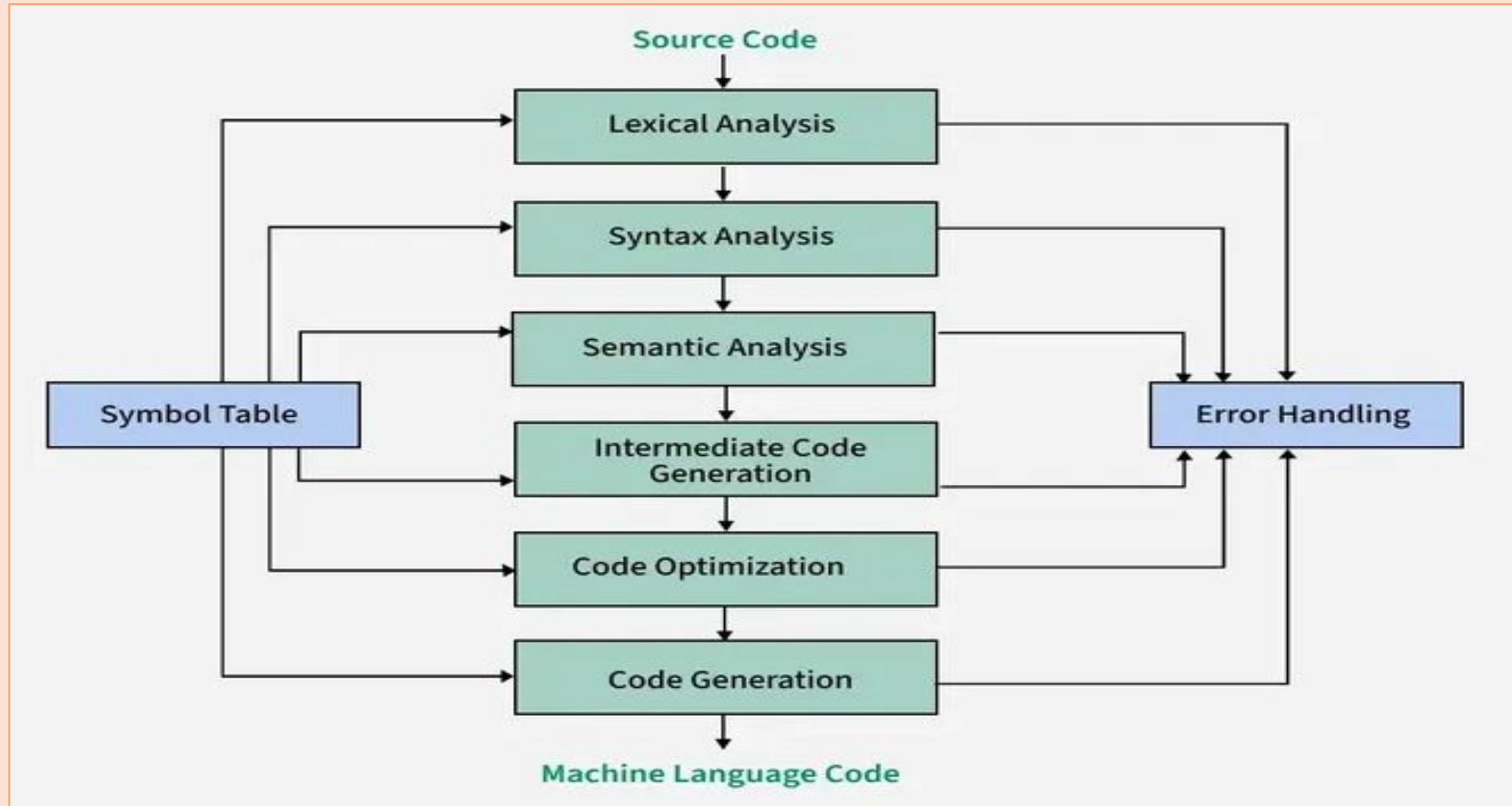
❖ **Functions**

- ☐ Converts source code into machine code.
- ☐ Detects and reports errors.
- ☐ Improves program efficiency.
- ☐ Generates executable files.

❖ **Why are compiler phases needed?**

A compiler divides its work into several phases so that each phase performs one specific task accurately and efficiently.

OVERALL PHASES OF A COMPILER



PHASE 1: LEXICAL ANALYSIS

❖ Purpose

The lexical analyzer reads the source program character by character and groups them into meaningful tokens.

❖ Input: `int total = a + b * c;`

❖ Responsibilities

- ☐ Removes spaces and comments.
- ☐ Identifies keywords, identifiers, operators and constants.
- ☐ Generates tokens.
- ☐ Detects lexical errors.

❖ Output

1. Token Stream

❖ Tokens Produced:

Lexeme	Token	Description
int	KEYWORD	Data type keyword
total	IDENTIFIER	Variable name
=	ASSIGN_OP	Assignment operator
a	IDENTIFIER	Variable
+	PLUS_OP	Addition operator
b	IDENTIFIER	Variable
*	MULTIPLY_OP	Multiplication operator
c	IDENTIFIER	Variable
;	SEMICOLON	Statement terminator

PHASE 2: SYNTAX ANALYSIS

❖ Purpose

Checks whether the sequence of tokens follows the grammar rules of the programming language.

❖ Example

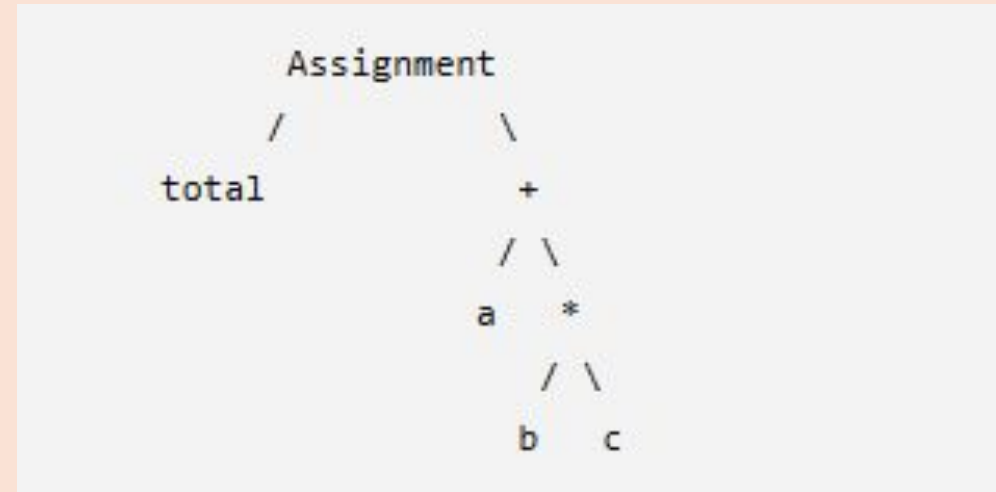
`int total = a + b * c;`

❖ Responsibilities

1. Builds Parse Tree
2. Detects syntax errors
3. Verifies program structure

❖ Output

Parse Tree / Syntax Tree



PHASE 3: SEMANTIC ANALYSIS

❖ Purpose

Checks whether the program is logically meaningful. / parse tree

❖ Example

```
int a;
```

```
a = "compiler";
```

Error

Cannot assign a string to an integer variable.

❖ Responsibilities

- ☐ Type checking
- ☐ Variable declaration checking
- ☐ Function parameter checking
- ☐ Scope verification

❖ Output

Semantically Correct Program

REMAINING COMPILER PHASES

❑ Intermediate Code Generation

Converts the source program into an intermediate representation.

Example (int total = a + b * c;)

t1 = b * c

t2 = a + t1

total = t2

❑ Code Optimization

Improves the intermediate code by making it

Faster

Smaller

More efficient

Example

t1 = b * c

total = a + t1

❑ Code Generation

Converts optimized intermediate code into machine code.

Example

MOV R1, b

MUL R1, c

MOV R2, a

ADD R2, R1

MOV total, R2

Output:

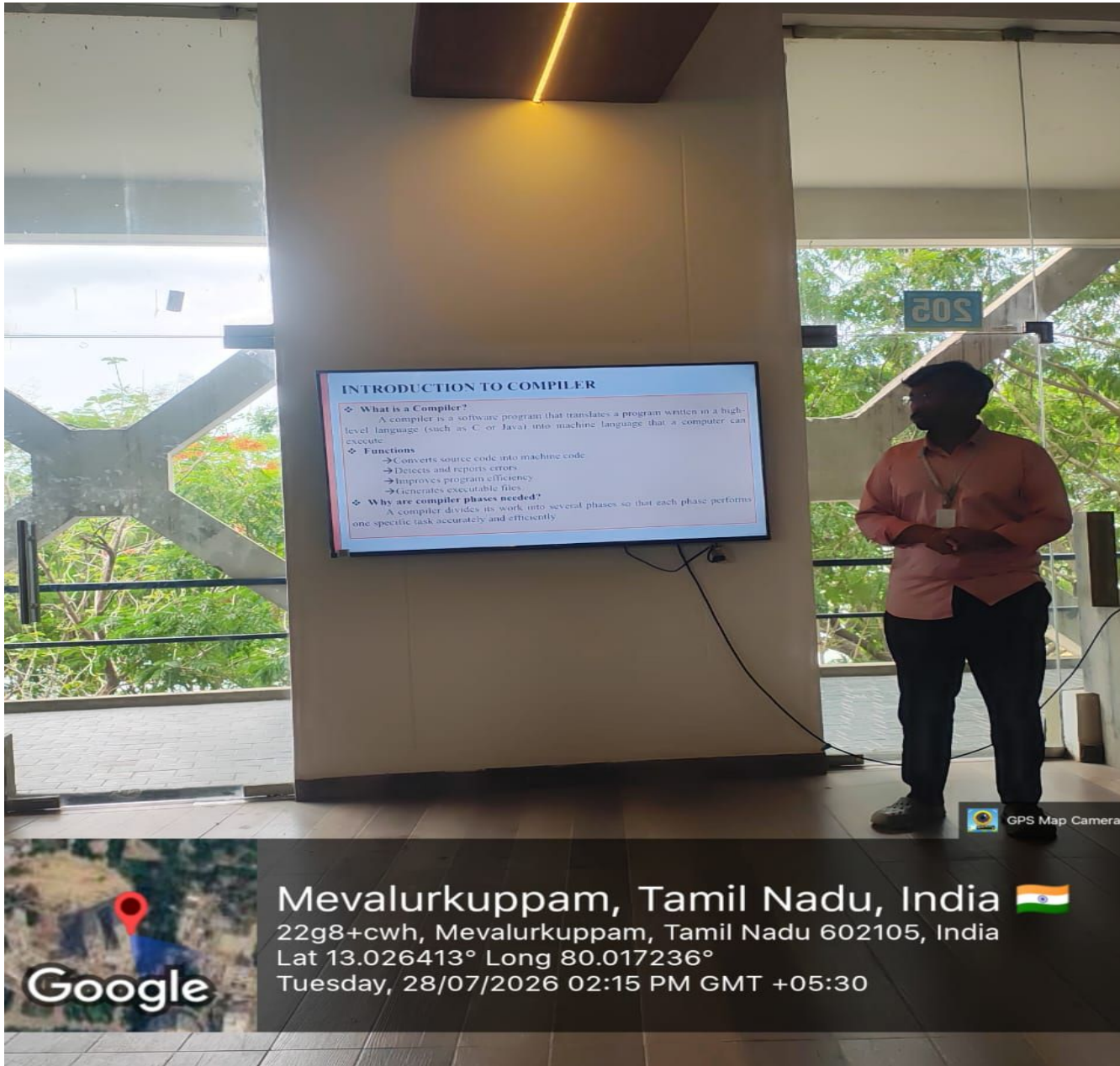
Executable Machine Code

CONCLUSION

- A compiler converts high-level programs into machine code.
- The compilation process is divided into multiple phases.
- Each phase performs a specific task independently.
- Sequential execution improves accuracy and maintainability.
- Optimization produces faster and efficient executable programs.

Key Takeaways

- ✓ Lexical Analysis → Generates Tokens
- ✓ Syntax Analysis → Checks Grammar
- ✓ Semantic Analysis → Checks Meaning
- ✓ Intermediate Code Generation → Creates IR
- ✓ Code Optimization → Improves Performance
- ✓ Code Generation → Produces Machine Code



INTRODUCTION TO COMPILER

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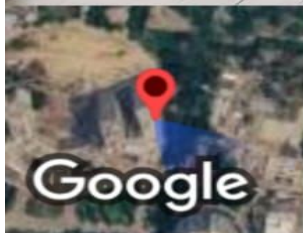
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Mevalurkuppam, Tamil Nadu, India 🇮🇳

22g8+cwh, Mevalurkuppam, Tamil Nadu 602105, India

Lat 13.026413° Long 80.017236°

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THANK
YOU